



Power Play: Investing in Energy Infrastructure for the AI Era

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This report includes contributions from the following Deutsche Bank research analysts: Brett Ryan (econ), Ross Seymore (semiconductors), Benjamin Soff (data center REITs), Michael Linenberg (airlines), Edison Yu (autos and space), Scott Deuschle (aerospace and defense), Nicole DeBlase (multi-industry and electrical equipment), Corinne Blanchard (clean tech), and Chris Robertson (LNG infrastructure and maritime transport).

The world is facing a significant energy deficit. This was true before AI and the war in Iran exacerbated the shortfall. The kinds of investments underway today will shape the future of the energy landscape and its ability to support an evolving tech-forward economy. Businesses are responding through legacy and innovative ways both to alleviate current constraints and to scale up for the long run.

To address immediate constraint issues, companies are sidestepping grid backlogs by offering readily dispatchable power sources and modular units. As the grid expands to meet growing demand, multi-industry and electrical equipment companies are enabling utilities to expand generation, transmission, and delivery. Renewables cannot be overlooked as a scaling solution as the AI era exponentially increases the demand for energy. Solar in particular is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems, these technologies can provide the kind of around-the-clock, predictable power that AI infrastructure requires. Commercial nuclear is another important component of the longer-term energy supply makeup, with the near-term focus on extending the lifetime of existing plants while looking ahead to longer-term expansion. All that said, even as the grid scales significantly and the mix of energy supply diversifies, natural gas is likely to remain a primary and backup source of power.

Authors

Laura Daniels

Deputy Associate Director
of Equity Research

Josh Ray

Associate Director
of Equity Research

Matthew Barnard

Head of US Company Research



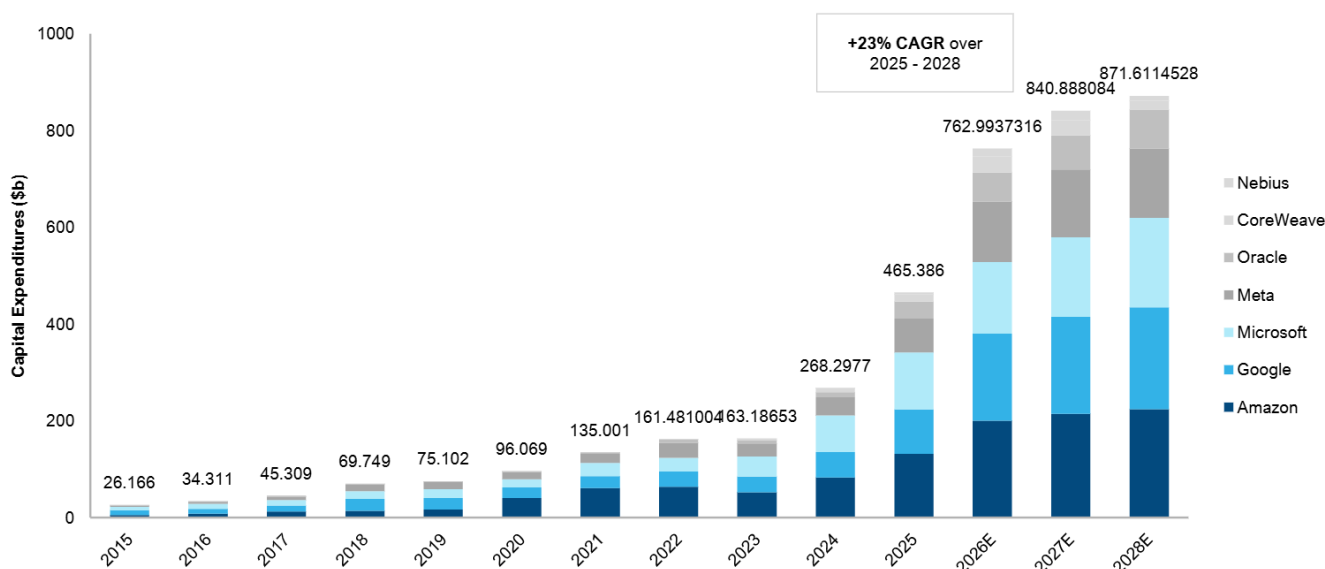
The Macro and Demand Landscape

The grid is under significant pressure. US energy generation hit a record 4.43 terawatt hours in 2025, according to the US Energy Information Administration; generation will likely break that record again this and next year due primarily to large data centers. Energy demand driven by AI data centers globally is expected to quadruple over the next four years, according to the International Energy Agency. To get a sense of how great this demand is and will be in the coming years, we can look to cloud-related investments, growth in data centers and AI workloads, and energy prices.

One of the most telling data points has been cloud-related investments. Our Semiconductors Research Team has tracked a decade of data-center-related investments via capital expenditures (capex) at several major cloud service providers to finance the setup of physical data centers as well as the electronic equipment needed to fill them. The data shows that the current buildout is unlike any other. Capex has surged over the last three years, rising from \$161 billion in 2022 to \$465 billion in 2025. This reflects a compound annual growth rate of +42%, a jump up from comparable three-year rates of +29% (2019-2022) and +30% (2016-2019).

Spending is expected to step up even further to \$762 billion (+64% year-over-year) in 2026. If that happens, this would mark the third year in a row that cloud-related capital investments have risen more than 60% year over year. Prevailing estimates also expect growth to slow after this growth spurt, but as we saw throughout 2025, consensus tends to underestimate the amount of spending in latter years, so future spending could well exceed today's expectations.

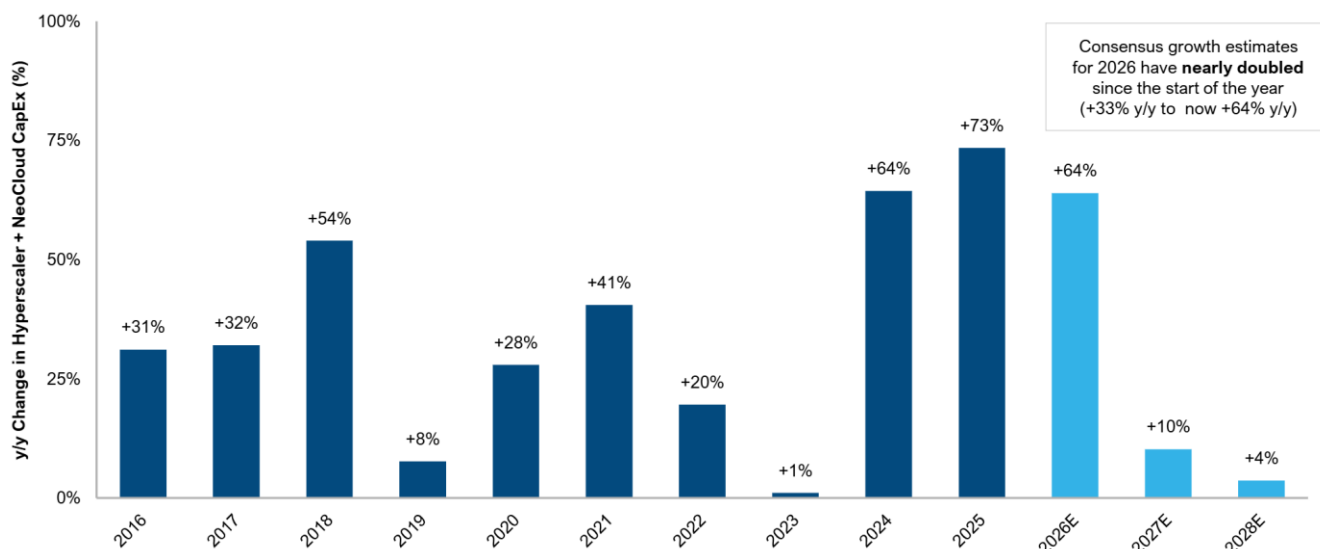
Figure 1: Aggregate Cloud Capex (2015 - 2028E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.



Figure 2: Annual Growth in Aggregate Cloud Capex (2016 - 2018E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data above is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.

Data center AI workloads paint a similar AI-driven energy demand picture. Data centers represent about 5% of total US electricity usage today, according to estimates from BloombergNEF and the Department of Energy. This will likely expand with the adoption of AI, as well as with continued growth in the digital economy.

Based on midrange estimates from McKinsey, total global data center capacity demand is set to increase from 82 gigawatts in 2025 to 220 gigawatts in 2030, amounting to a 22% annual growth rate. This estimate sees generative AI workloads growing at 29% on average, plus 11% from non-AI workloads. Generative AI workloads could grow to 35% of total global data center activity by 2028, up from a mid-single-digit percentage as of 2023, according to estimates from Boston Consulting Group.

AI workloads require greater power density (essentially the amount of electricity a system needs per square foot) and more intensive liquid cooling, driving up energy consumption. Data center companies like Digital Realty (DLR) and Equinix (EQIX), for example, are already seeing these trends play out in their businesses. Digital Realty recently said it is rolling out higher-density deployments with pre-installed liquid cooling to meet AI demand. Equinix has noted that 60% of its larger deals in the last quarter of 2025 were driven by AI, up from 50% earlier last year. The power density for these deals, according to Equinix, is 33% higher than for non-AI deals.

These factors exacerbate concerns around US electricity prices and grid capacity, which were already mounting even before the war in Iran elicited a spike in global energy prices. After an extended period of stable growth in the US – over the last decade, electricity output grew at about a 0.7% annualized rate – data center electricity demand estimates are raising concerns about a knock-on



impact to household utility bills as well as grid capacity. The energy problem is just as acute for central bankers. Traditionally, Federal Reserve policymakers would tend to look through the temporary inflationary impact of an energy shock (see US Economic Perspectives: Impact of oil price shocks a "frack"-tion of what they once were). However, the backdrop to the latest energy shock is one in which inflation has persistently exceeded the Federal Reserve's 2% goalpost for the better part of the last five years. This raises the risk that inflation expectations could become unanchored. In short, the energy supply outlook carries implications across a breadth of economic issues, from interest rates to technology stocks.

These concerns could result in policy favorable to companies increasing energy supply, as energy prices will likely be top of mind for voters ahead of the midterm elections this November. As we recently outlined, historically, the relationship between gasoline prices and consumer energy spending suggests that a 1 cent rise in gasoline prices (on a sustained basis) increases spending on energy goods and services. This thus reduces income that can be spent on non-energy items – by \$1.15 billion, amounting to a sizable "energy tax." Mapping the impact of potential energy price shocks to consumers, we find that at a sustained \$100 per barrel oil price, the projected tax benefits to consumers from last year's tax legislation (the "One Big Beautiful Bill Act") would still exceed the drag from the energy tax increase. However, at \$150 per barrel, the implied energy tax increase would more than wipe out the tax cuts to consumers from that legislation and present a more serious threat to the outlook for consumer spending.

The sum total of this is that supply is already struggling to keep up. Wait times to connect a data center to the grid are exceeding four years (based on comments from Google representatives and other industry reports). Efforts are underway to increase supply, including expanding grid capacity, as well as AI players preparing to build their own sources of power that operate "behind-the-meter" (developed on the customer's side and not delivered through the public grid). Capex into investor-owned electric utilities has been steadily increasing and is projected to climb to nearly \$250 billion in 2029 (according to the Edison Electric Institute). The White House's introduction in early March of a Ratepayer Protection Pledge urging AI companies to build or buy additive energy further highlights the race to bolster energy supply to meet AI needs. While the exposure across regions will vary, the conflict in Iran will add another layer of constraint to energy supply, as 20% of global oil and 20% of LNG pass through the Strait of Hormuz. Companies are looking to fill the gaps in energy supply over a spectrum of problem sets solving for immediate constraints, for scale, and for consistency.



Solving for Constraint: Readily Deployable Power

The most direct response to AI-driven power constraints is coming from companies that enable data center operators to bring electricity online without waiting for grid expansion. These companies are delivering dispatchable on-site or near-site power. While there are existing manufacturers providing these kinds of products, there are an increasing number of companies adapting existing technology to enter this AI-adjacent space. FTAI Aviation (FTAI), which traditionally operates in the airline industry, and BorgWarner (BWA), an auto industry supplier, are examples of this kind of shift.

FTAI Aviation launched FTAI Power in December 2025 with the aim of converting the company's CFM56 engines into gas turbines that can deliver energy to data centers. The engine is prevalent, with enough in operation globally that one takes off every two seconds somewhere in the world. FTAI Power will offer its aeroderivative turbine "FTAI Mod-1," a mobile, trailer-mounted conversion unit that can be delivered and installed in under two weeks. As a 25-megawatt unit, it is smaller, which allows for flexibility in stacking and growing data centers. Deliveries will likely begin around December 2026, with approximately 100 units in production per year starting in 2027.

BorgWarner is taking a similar approach. The company is an auto industry supplier that designs and manufactures powertrain systems but will partner with Turbocell (a subsidiary of the AI-optimized data center infrastructure company Endeavour) to co-develop an integrated, modular power generation system. This system would provide primary and backup power for AI data centers and would serve as a "bridge" solution, meaning data center operators can install it quickly for power now but can keep it on to provide additional power even as longer-term energy infrastructure is built up.

Modular power generation systems like this one are designed with the high and often fluctuating power demands of AI computing in mind. Modular products also comprise standardized, interchangeable units that are more flexible in applications than the custom builds commonly found in legacy systems. These modular units, as well as the integrated battery systems and electronic components, can be scaled up or down based on specific data center needs.

Beyond quick-install energy solutions, companies further up the supply chain are adapting to the energy demanded by growth in AI. Woodward, for example, is an aerospace-industrial hybrid company whose power generation products have historically been focused on the industrial gas turbine market, but it is now broadening into supplying components for reciprocating engines and marine engines being repurposed as behind-the-meter power solutions. |



Solving for Scale: Grid Expansion and Infrastructure Development

Meeting AI-driven electricity demand forecasts will require system-wide capacity and infrastructure expansion. Investments are happening simultaneously but across pathways that range from near-term grid expansion and enablement platforms, near- and longer-term renewables, and long-duration commercial nuclear supply. In terms of grid expansion, efforts span both generation capacity and transmission and distribution (T&D) infrastructure to accommodate higher loads and load density, more stringent reliability requirements, increasingly complex grid architectures, and longer-duration power needs associated with AI workloads. GE Vernova (GEV), for example, manufactures combined-cycle and peaker gas turbines (currently the most common source of baseload power in the US) and also provides high-voltage T&D equipment including transformers, high-voltage direct current (HVDC) solutions, circuit breakers, switchgear, power-sending devices, and grid automation systems. Some of its products also supply the renewable and commercial nuclear sources of power.

As existing grid infrastructure faces a critical deficit, it is impossible to ignore renewables as a scaling solution. Although US onshore wind investment faces challenges due to unfavorable policy headwinds, solar, alongside other renewables, is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems (BESS), these technologies can meet the 24/7 power needs of AI-focused data centers. There are indications that as energy demand driven by AI increases, so too do investments in renewable energy sources. A recent study from the University of Zurich, for example, found that solar and BESS have been the primary drivers of increased energy investment in regions with pre-existing data center capacity (based on ChatGPT infrastructure in the US). Increased investments in gas play a role too, but these have been offset by declines in grid-connected gas plants. These developments are currently based on investment decisions, but policy drivers could add further impetus should the environmental toll stemming from AI usage take an increasingly prominent place in political discourse.

Solar energy is well suited to the energy demand landscape due to the technology's fast scalability, low to zero marginal costs, and favorable levelized cost of energy (LCOE). As a result, the sector is seeing an increasing number of power purchase agreements (PPAs) with large technology players, including Amazon (AMZN) and Google (GOOGL). These arrangements are also conducive to a buildout of development projects that are supporting equipment suppliers.

Clearway Energy (CWEN), for example, operates energy generation assets in the US, with a specific focus on clean tech. It is positioning itself to serve the growing demand for power in the US driven by hyperscalers and data centers. The company has announced 2 gigawatts of PPAs for these projects, with possibilities for additional volumes. Additionally, it has outlined 8-13 gigawatts of capacity via co-located data center complexes, utilizing a mix of renewables and storage in addition to gas.

The scale-up of renewables also depends on component and equipment suppliers. First Solar (FSLR), for example, is a US-based solar technology



company that designs and manufactures advanced thin-film modules (a kind of solar photovoltaic panel) for utility-scale projects. Nextpower (NXT), meanwhile, is an example of a clean tech equipment and systems provider, selling trackers (the mechanical systems that hold and rotate panels to follow the sun), software, electrical systems, and the foundations upon which solar technology is mounted. Solar electrical equipment also includes inverters, which convert solar-generated direct current electricity into distributable alternating current electricity delivered to both residential and commercial customers.

Commercial nuclear is another important component of longer-term energy supply, with small modular reactors (SMRs) a key part of the development pipeline. Businesses are focused on extending the lifespan of current reactors while also preparing for the possible buildout of new ones. In the near term, the priority is the maintenance, repair, and life extension of existing nuclear infrastructure. There are 54 nuclear power plants in the United States (amounting to 94 nuclear reactors), according to 2024 data from the US Energy Information Administration. These produce a net generating capacity of nearly 97 gigawatts and make up about 20% of US energy usage.

Maintenance of these plants requires the replacement and upgrading of critical reactor components, modernization of control systems, and other projects that allow current reactors to operate safely and consistently for longer periods. Suppliers such as BWX Technologies (BWXT) and Curtiss-Wright (CW) provide operating components and systems for nuclear plants, including reactor hardware, fuel-related technologies, pumps, valves, and control systems.

At the same time, the industry is preparing for potential new build activity, particularly around advanced reactor designs and SMRs. Development work spans reactor components, fuel technologies, and digital and mechanical systems designed for newer reactor platforms. BWX Technologies' work on advanced fuel and reactor components and Curtiss-Wright's involvement in systems and controls applicable to newer reactor designs are examples of how suppliers are positioning for a more modular and scalable nuclear build-out over time.



Solving for Consistency

At the core of scaling AI infrastructure is the need for stable, reliable, and low-cost power. While wind and solar power generation will undoubtedly be part of the mix, natural gas can be expected to continue to play a sizable role as both a primary source of power and a backup source of power to intermittent power sources such as wind. Even with significant renewable penetration, the grid will still require backup generation capacity. Natural gas, in our view, will remain the likely fuel selected for this purpose, due to its global availability (via pipeline or liquefied natural gas imports), energy density, price, and ability to be transported and stored on site for behind-the-grid power. Natural gas also avoids many of the political hurdles that constrain other energy sources. It does not face the regulatory and political hurdles of traditional nuclear and SMRs, nor is it falling out of favor in Western markets to the same degree as coal. As such, it is likely that natural gas will continue to be in demand for both baseload and backup power generation.

The Bottom Line

Projected energy demand tied to the growing AI ecosystem and the broader digital economy is rising faster than grid capacity and interconnection timelines can adjust. Businesses are responding to address immediate constraint issues, questions of scale, and the need for consistent primary and backup generation. This will shape not only the trajectory of the AI buildout, but have reverberating impacts across the broader economy, its competitiveness, and critical infrastructure.



Appendix 1

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